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1. Introduction

1.1 Background and Context

These days, artificial intelligence and machine learning play key roles in how decisions are made using data - from hospitals to banks, schools to online shopping platforms. Because data keeps expanding in size and complexity, it matters more than ever that such systems can organize, understand, and draw conclusions from information. Instead of just processing numbers, they must make sense of patterns to guide both automated tools and people making choices. How well data is represented and analyzed directly shapes what these technologies can do.

What lies at the core of artificial intelligence? Knowledge Representation shapes how information from specific areas gets coded so machines can process it logically. Semantic maps, frame structures, logical models, or shared frameworks help define links, rules, and situational context within data. Instead of relying solely on predefined facts, today's learning algorithms detect trends by analysing vast datasets using math-driven processes. These two paths - structured understanding and pattern extraction - often work better together than apart. Systems tackling real-world complexity gain strength when both forms are balanced thoughtfully.

1.2 Aim and Objectives

This assessment takes a close look at how knowledge is structured within artificial intelligence, using actual data to test methods from both supervised and unsupervised learning. While one part focuses on applying these techniques in practice, another explores their underlying principles through hands-on analysis. Each step connects theory with measurable outcomes, showing how concepts shape real computational behavior. Rather than staying abstract, the work grounds ideas in concrete model performance.

The specific objectives of this assessment are to:

Looking at how information is structured in machines reveals much about their thinking patterns. Different models shape the way systems draw conclusions. Some methods allow faster choices, others support deeper understanding. The design influences accuracy and speed differently across tasks. Structure affects what a system can explain after deciding. Certain formats work better when uncertainty exists. How data connects inside determines possible outcomes. Each approach carries trade-offs between flexibility and clarity. Not every technique suits all kinds of problems equally well.

Identify a real-world problem that can be addressed using machine learning techniques.
Begin by choosing a suitable algorithm - either supervised or unsupervised - for the given dataset. Then proceed to run it on the data, ensuring alignment between method and task. Execution follows selection without deviation from predefined objectives. Results emerge only after correct pairing of technique and information structure.
Evaluate model performance using suitable quantitative and qualitative measures.
Reflect on the challenges, limitations, and ethical considerations associated with applying artificial intelligence techniques in practice.

1.3 Brief Overview of Structure

The layout follows a path between ideas and hands-on work. One part digs into how machines handle knowledge, looking at methods used, what they allow systems to figure out, then where they fall short when things get messy. Moving forward, another section walks through actual steps taken - starting with framing the task, choosing information sources, cleaning inputs, building models, checking outcomes. A later segment pauses to think about difficulties faced, whether any of it holds up outside labs, moral questions raised, along with ways it might be done better next time. At its close, the document wraps up by pulling together what stood out most from the effort described earlier.

2. Theoretical Analysis of Knowledge Representation

2.1 Overview of Knowledge Representation in Artificial Intelligence

What lies behind how machines understand things is shaped by Knowledge Representation (KR), which structures information clearly for artificial systems to process. It organises facts, rules, and connections in ways that allow software to draw conclusions or take actions logically. Through KR, unprocessed data transforms into meaningful patterns that support decision-making. Rather than relying only on statistical trends, it highlights defined logic, dependencies, and boundaries of a subject area. The strength comes not from volume but from clarity - how well concepts link and interact inside a system.

Because artificial intelligence depends on how knowledge is shaped, picking the right format changes how well systems grasp data or adapt. Logic-based symbols once led the way, guiding decisions through clear rules and defined structures. Now, even though numbers and patterns drive most current models, organizing facts in readable forms still matters when

clarity and reliability count. So knowing which methods work - and why they matter - shapes whether smart tools can handle messy real-world demands.

2.2 Key Knowledge Representation Techniques

Starting with basic structures, researchers built various ways to capture how people organise what they know. Nodes linked by lines form semantic networks, mapping ideas through connections rather than isolated facts. These links show how one concept relates to another, building meaning through structure instead of words alone. Though straightforward when drawn out, such models struggle with exact interpretations due to missing rule-based definitions. Because of that gap, drawing firm conclusions becomes difficult even if the layout looks clear.

What makes frames stand out is how they shape information using fixed patterns, where each template holds details about things or events via labeled properties. These structures work well when capturing typical scenarios and common expectations people carry. Unlike basic semantic nets, such systems allow traits to pass down across levels, while also managing unusual cases without breaking rules. Yet, when ideas shift rapidly or stay too vague, this approach can fall short.

Starting with clear symbolic structures, propositional and first-order logic form a strict mathematical foundation for encoding knowledge. Because of their structured syntax, these approaches support exact conclusions via established deduction techniques. Despite delivering reliable and provably correct outcomes, such frameworks often demand high processing resources. When faced with ambiguous data or missing details, they tend to struggle due to limited adaptability.

2.3 Impact on Reasoning and Decision-Making

How knowledge is structured shapes the way artificial intelligence processes information and reaches outcomes. When using logical formats, machines follow strict rules to produce answers that must be true if the starting points are correct. On the other hand, network-like structures connect ideas through relationships, leading to inferences based on similarity or context rather than certainty. These differences influence both speed and reliability in problem solving.

When clear frameworks like ontologies help understanding and uniformity, they can limit choices during unfamiliar or unclear scenarios. On the other hand, looser forms of representation offer flexibility, yet risk confusion or mismatched logic. So, how well reasoning

works relies heavily on both the method used - and the way information is shaped and arranged inside the system.

2.4 Challenges, Limitations, and Failure Scenarios

When conditions shift quickly, stored information risks falling behind. Systems built on fixed rules struggle as situations evolve. Take ontology models - these rely heavily on human oversight to stay current. Without regular adjustments, conclusions drawn might no longer hold true. What works today could become unreliable tomorrow.

When knowledge bases expand in size and complexity, scalability becomes a growing concern. Logic-driven methods often struggle under heavier computational loads. Inference may take longer or require simplified models instead. These simplifications sometimes lower overall precision.

Frequently, known methods of knowledge representation find it hard to handle unclear or unsure data. Since people's understanding tends to be partial or based on likelihoods, fixed logic models fall short - they rely on precise, solid truths instead. These structured formats work fine under stable conditions yet break down easily with messy, conflicting, or fuzzy inputs. Take strict rule-driven setups: they might reach wrong outcomes whenever expected patterns shift or unusual cases stay unmentioned.

What holds back KR methods shows up clearly when they face actual tasks outside controlled settings. Because of this, blending them with learning-based systems becomes less a choice and more a necessity. Real situations rarely fit clean logical structures. Without adjusting their design or combining forces elsewhere, these techniques struggle to keep pace. The gap between theory and practice grows unless updates happen. Their usefulness fades if left unchanged.

3. Practical Implementation

3.1 Problem Definition and Domain Context

3.1.1 Domain Description

Despite generating large volumes of interaction data - such as time spent on pages, paths taken through sites, when visits occur, who the visitors are - online shopping platforms face challenges in interpreting this information clearly. Behaviour rarely follows predictable patterns; actions shift abruptly, differ widely between users, complicating straightforward analysis. While details like click sequences or session lengths offer clues about interest levels, turning them into useful strategies demands more than summary metrics. Firms now lean on machine learning tools to detect subtle trends within chaotic user activity, especially as market pressure grows. These technologies help uncover hidden structures in how people explore products, respond to layouts, make purchases. With sharper insight comes improved targeting, smoother experiences, higher completion rates at checkout. Such methods have become common practice across digital retail spaces aiming to stay responsive and competitive (Laudon & Traver, 2021).

3.1.2 Problem Statement

Predicting if someone browsing online will actually buy something sits at the heart of e-commerce analysis - this task comes paired with identifying how various visitor behaviours set people apart. Although plenty of visits end without a sale, those who skip buying aren't all alike; their goals, attention levels, and potential worth later on can vary widely. Spotting signs of purchase interest early allows stores to step in thoughtfully, perhaps by suggesting relevant items or offering tailored deals. Still, knowing what might happen tells us little about why actions unfold differently across individuals. When companies overlook the hidden groupings within customer activity, they often fall back on one-size-fits-all tactics, which tend to miss the mark for varied audiences (Aggarwal, 2018).

3.1.3 Justification for algorithm pair selection

This research tackles the issue using both supervised and unsupervised methods together. Because logistic regression works reliably for yes-or-no outcomes, it serves as the main tool for forecasting whether a purchase will happen - its clear output helps explain how decisions are made (James et al., 2021). For uncovering hidden patterns in user activity, k-means clustering takes centre stage, sorting browsing sessions into meaningful clusters based on

similar actions. Since k-means relies on measuring distances between data points, it performs well when mapping out habits within vast streams of behavioural records (Han, Kamber, & Pei, 2020). When used alongside one another, prediction gains depth through group insights, painting a fuller picture of customer tendencies while aiding practical choices and long-term planning in online retail settings.

3.2 Dataset Selection and Justification

3.2.1 Dataset source and description

Available through the UCI Machine Learning Repository, the data comes from an analysis of online shopper behavior. Each entry stands for one browsing session on a retail site. Instead of grouping actions, it records distinct visits by users. Attributes mix numeric values with category labels, capturing how people navigate pages. Time spent on various sections appears alongside counts of those interactions. When applicable, details about who visited show up too. A key outcome tracked is transaction completion. That result - yes or no - is labeled as Revenue per session. Source: Sakar et al., 2019.

3.2.2 Suitability for supervised learning

Because the outcome splits into two clear categories, the data fits naturally with methods that sort observations into classes. Features tied to user actions carry useful signals about buying interest, so patterns in behaviour during a visit can inform predictions of whether a sale happens. These forecasts matter in online retail settings - spotting visits with strong intent opens paths to timely responses and adjustments aimed at boosting completed purchases (James et al., 2021).

3.2.3 Suitability for unsupervised learning

Beyond its role in supervised settings, this collection includes diverse behavioural indicators useful for exploratory methods. Because they capture subtle differences across interactions, these elements help expose hidden structures within session records, independent of transaction outcomes. Patterns emerge when grouping similar actions, showing how people navigate differently, invest varying attention, or signal interest in unique ways - insights often missed by forecast-driven approaches (Han, Kamber, & Pei, 2020).

3.2.4 Ethical and data quality considerations

Even though personal details are removed, the data could still pose privacy concerns because patterns might reveal private behaviours when viewed incorrectly. What makes things tricky is that most records show visits without a buy, creating an uneven spread that may skew how well models predict real outcomes. Though identities are hidden, using results carelessly might lead to unfair treatment of certain groups. A balanced approach matters - especially since reliability depends on handling both fairness and accuracy throughout analysis.

3.3 Data Preprocessing and Feature Engineering

3.3.1 Data Cleaning and Handling Missing Values

Starting off, a basic check on data quality helped spot any gaps and judge how complete the Online Shoppers Purchasing Intention dataset really was. It turned out most variables had little to no missing entries, which suggests the information is quite dependable. Because of this, there was no need to fill in blanks or drop records; every case stayed in place for further examination. That way, natural shopping behaviours remained intact, untouched by artificial changes that might skew results. Looking closer at the Revenue outcome, its spread showed a clear tilt toward sessions without purchases. Such a pattern lines up with what happens on online stores regularly, where most visits do not lead to buying. That uneven split played a role later when judging how well models performed.

```

Missing values (top): Administrative      0
Administrative_Duration      0
Informational                 0
Informational_Duration       0
ProductRelated               0
ProductRelated_Duration     0
BounceRates                  0
ExitRates                    0
PageValues                   0
SpecialDay                   0
Month                        0
OperatingSystems             0
Browser                      0
Region                       0
TrafficType                  0
VisitorType                  0
Weekend                      0
Revenue                      0
dtype: int64

```

Figure 1 : Missing value check

3.3.2 Feature Engineering

Starting with the data setup, feature engineering shaped the dataset for use in both labeled and unlabeled modeling approaches. Before any training began, Revenue stood apart from predictors to keep outcomes distinct. When turning to clustering, that outcome stayed out entirely - keeping patterns free from influence by known results. Details about user sessions and traits appeared as categories, left intact but ready for conversion into numerical form. By handling them this way, subtle behavior signals remained visible without clouding how models could be understood later.

```

Target value counts:
Revenue
False    10422
True      1908
Name: count, dtype: int64

```

```

Numeric features: 14
Categorical features: 3

```

Figure 2 : Feature Engineering

3.3.3 Normalisation and Scaling

Scaling adjusted the range of numeric data so values like page visits or timing metrics did not dominate due to size alone. Because models relying on distances weigh larger numbers more heavily, transforming these inputs helped balance their impact. One method reshaped each feature to fit a standard frame, reducing bias from raw measurement scales. With uniform dimensions, comparisons between individuals became more reliable. Predictions and groupings alike benefited from this level of consistency.

```

numeric_transformer = Pipeline(steps=[
    ("scaler", StandardScaler())
])

categorical_transformer = Pipeline(steps=[
    ("onehot", OneHotEncoder(handle_unknown="ignore", sparse_output=False))
])

preprocessor = ColumnTransformer(
    transformers=[
        ("num", numeric_transformer, numeric_features),
        ("cat", categorical_transformer, categorical_features),
    ],
    remainder="drop"
)

```

Figure 3 : Scaling of numeric values

3.4 Model Development

3.4.1 Supervised Model Implementation (Logistic Regression)

Following the setup of training data, prediction relied on Logistic Regression for distinguishing purchase outcomes from non-purchases. Before splitting, variables were grouped - some became inputs, one served as outcome. When needed, logical values shifted into numbers so scoring methods could process them correctly. Inputs mixed countable measures with labelled groups, capturing actions during visits along with surrounding conditions.

Stratified by the outcome, the data split into train and test sets kept class proportions intact. Consistency in preparation steps came from structuring operations sequentially. For numeric columns, values shifted to zero mean and unit variance through scaling. Categorical inputs took shape as binary flags - unknown categories set aside during encoding. Within that framework, logistic regression learned patterns directly from adjusted features. Process flow stayed unified, linking cleanup and learning in one chain. When dealing with uneven class distributions, adjusting class weights helped balance the training process. Because the model delivered predicted classes along with likelihood scores - when those were accessible - it allowed performance assessment without relying on fixed thresholds, using ROC curves to guide interpretation.

```

--- Logistic Regression Evaluation ---
Accuracy : 0.8500
Precision: 0.5107
Recall   : 0.7487
F1-Score : 0.6072
ROC-AUC  : 0.8962
  
```

Figure 4 : Logistic Regression Evaluation

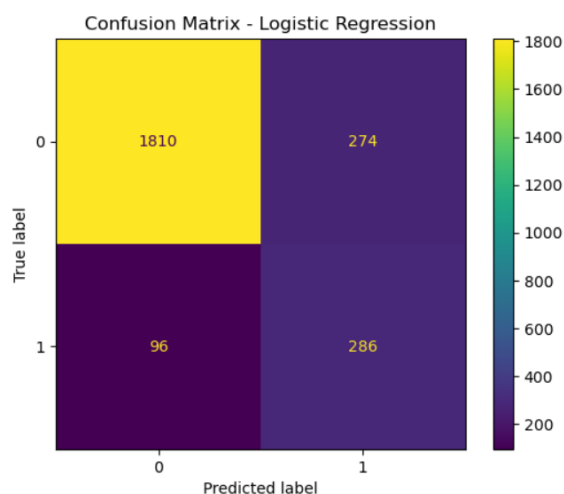


Figure 5 : Confusion Matrix

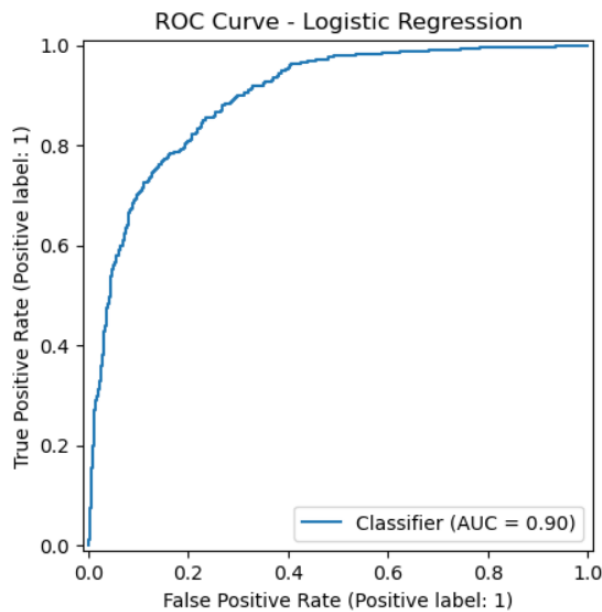


Figure 6 : ROC Curve

3.4.2 Unsupervised Model Implementation (K-Means Clustering)

A pattern-finding step relied on K-Means to detect hidden categories of shopping visits by matching how users behaved. To build clusters, a cleaned data table came from applying an identical transformation process across all features - except revenue, which stayed out to avoid influencing results unfairly. Because distances shape K-Means outcomes, rescaling numbers and converting labels into codes kept each trait equally influential despite differing units.

A closer look at clustering began by testing several possible numbers of groups, checking how tightly data points grouped together using inertia. Instead of relying on just one measure, the analysis also brought in Silhouette Scores to weigh how clearly separated each group remained. Where clusters showed highest clarity through the silhouette lens became a top choice for k. This picked value shaped the final version of the K-Means setup. After fitting, every session received a label based on its matched group, then those tags joined the main dataset to help make sense later.

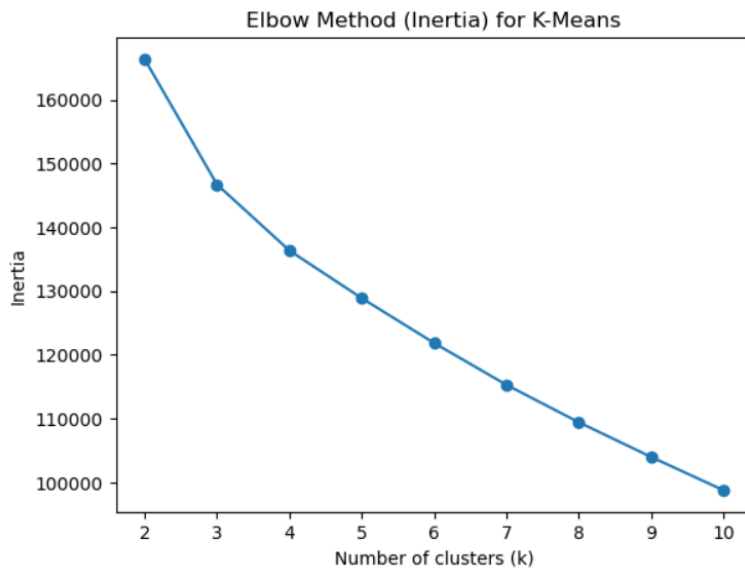
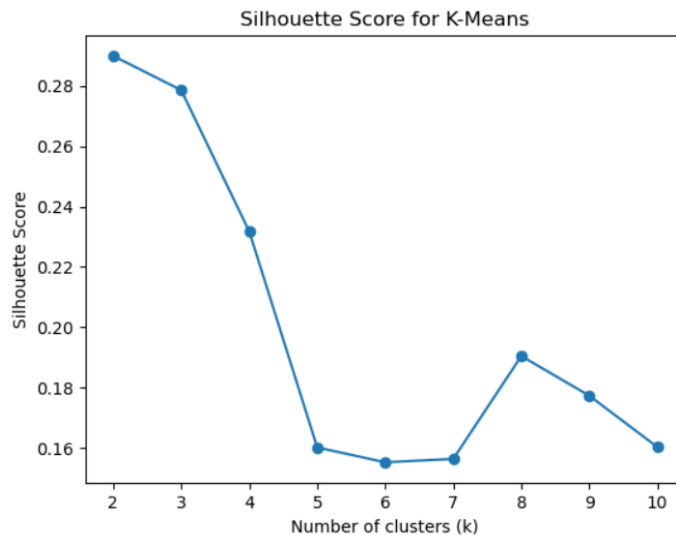


Figure 7 : Elbow Method



Best k by silhouette: 2

Figure 8 : Silhouette Score

3.4.3 Visualisation and Output Generation

Outputs needed for assessment and documentation came from the notebook during model creation. When classifying, results showed through a confusion matrix; if probabilities existed, an ROC curve also appeared. Clustering relied on Elbow and Silhouette visuals to guide group choices. Profile work followed, using size summaries and number-based averages per group - sometimes paired with category breakdowns or buying frequency by segment - to clarify meaning for decision makers.

3.5 Evaluation and Analysis

3.5.1 Supervised Evaluation Metrics

To judge how well the supervised learning model worked, several classification measures were applied - accuracy, precision, recall, and F1-score among them. Because the data showed far more non-purchasing than purchasing sessions, these particular indicators helped ensure a fair evaluation across both groups. Overall correctness was reflected through accuracy. Precision and recall together revealed whether the model could spot buying behavior while keeping mistaken predictions low. Instead of treating those two separately each time, the F1-score combined their balance into one value. Though useful alone, none of these numbers told the whole story without context.

Apart from numbers, the confusion matrix helped show how often predictions were right or wrong for each group. Seeing these patterns makes it easier to grasp where the model struggles. When probabilities existed, performance across decision levels was checked using ROC curves. The area under that curve gave a sense of separation strength between buy and non-buy moments.

3.5.2 Unsupervised Evaluation Methods

To check how well K-Means clustering worked, two approaches were combined: the Elbow Method alongside the Silhouette Score. Looking at shifts in within-cluster variance made it possible to apply the Elbow Method across different numbers of clusters, revealing where adding more brought little improvement. Using the Silhouette Score allowed measurement of how tightly data grouped together and how distinct those groupings remained, confirming that the chosen setup formed clear, valid clusters.

3.5.3 Comparative Insights

The analysis shows how supervised and unsupervised methods strengthen each other. Prediction at the session level comes from supervision, whereas patterns in actions emerge through grouping similar cases. One builds forecasts, yet the other uncovers hidden structures behind choices people make online. These paths, when used alongside one another, give a fuller picture - tying precise results to meaningful explanations without losing depth.

===== FINAL MODEL SUMMARY =====

Supervised Model: Logistic Regression

```

-----
Accuracy      : 0.8500
Precision     : 0.5107
Recall        : 0.7487
F1-Score      : 0.6072
ROC-AUC       : 0.8962
  
```

Unsupervised Model: K-Means Clustering

```

-----
Final number of clusters (k): 2
Best k by Silhouette Score : 2
Maximum Silhouette Score  : 0.2900
  
```

Cluster sizes:

```

Cluster
0      1866
1     10464
Name: count, dtype: int64
  
```

Average Revenue (Purchase Rate) by Cluster:

Revenue_Rate	
Cluster	
0	0.279743
1	0.132454

=====

Figure 9 : Summary of model comparison

4. Reflection and Discussion

4.1 Challenges Encountered

During deployment of both supervised and unsupervised approaches, difficulties emerged. The skewed distribution of classes posed a major hurdle - purchase events made up only a small fraction compared to non-purchase interactions. Because one group dominated numerically, the classifier tended to lean toward predicting that outcome more often, which could distort how well it actually performed when judged by accuracy alone. Adjustments came through assigning higher weights to underrepresented labels while fitting the algorithm; at the same time, performance scrutiny relied on alternative indicators instead of

relying solely on overall correctness rates. A different hurdle comes from mixed data types - numbers alongside categories - that stretch across uneven ranges. Handling this meant adjusting values through scaling, transforming labels into usable formats. Such steps helped balance influence during analysis, stopping any one variable from skewing outcomes too much. Distance-sensitive methods especially need this kind of preparation to work reliably.

4.2 Real-World Applicability

Using both labelled-data training and pattern discovery improves practical outcomes in online retail settings. When systems anticipate buying interest, they respond instantly - activating tailored discounts, guiding product suggestions, or alerting service teams during active visits. At the same time, grouping users by how they navigate reveals clear types of shoppers through their actions and involvement depth. Such groupings shape ad planning, interface layout adjustments, along with long-term client interactions via focused tactics. With both techniques applied, companies shift from just reviewing past activity to shaping future choices guided by insight.

4.3 Ethical Considerations

Even if personal details are absent, working with behaviour patterns still raises ethical questions. When some actions appear more often in data, predictions might tilt unfairly because of it. Outcomes shaped by such imbalances could nudge systems toward exclusion without careful checks. Decisions driven by forecasts about buying choices sometimes cross into uncomfortable territory, especially without oversight. Clarity in how conclusions are reached matters just as much as accuracy. Models that allow inspection, paired with honest assessment methods, keep real-world use grounded and fair.

4.4 Limitations and Future Improvements

Even though the method worked well, some drawbacks remain clear. Because Logistic Regression relies on linearity between predictors and log-odds, it might miss intricate human behaviour trends. In much the same way, K-Means expects clusters that are round and evenly spread - conditions rarely met in actual behavioural datasets. Moving forward, researchers might look into richer alternatives, like boosted trees for predicting outcomes or DBSCAN when grouping individuals by pattern similarity. Looking ahead, adding time-based patterns might clarify how users act. Another path involves testing the results with outside

data, which tends to support broader conclusions. What matters is whether these outcomes hold up when used live.

5. Conclusion

5.1 Summary of Findings

Looking into how people shop online, this work mixed ideas with real tests using artificial intelligence. Instead of just theory, it showed how setting up knowledge clearly helps machines think through choices. Yet those rigid methods struggle when things change fast around them. On the hands-on side, training models on past results worked well for guessing what users would buy. At the same time, letting algorithms find hidden structures exposed natural groupings in browsing sessions. Mixing both ways gives a fuller picture than either alone could offer.

5.2 Key Contributions

What stands out here is combining guided and unguided machine learning approaches into one system tested on actual online shopping data. Instead of treating prediction and grouping separately, the research shows their joint power in shaping daily operations alongside long-term planning. Another key point involves leaning on clear models and step-by-step validation - this keeps AI tools open to scrutiny while matching tech rigor with real company needs.

5.3 Final Remarks

Though limited to a specific setting, the methods shown here can apply elsewhere whenever human actions need forecasting. This work shows how blending theory about knowledge structure with hands-on ML builds solutions for actual challenges. Attention to cleaning data, choosing correct metrics, and recognising moral implications stands out across outcomes. Even if circumstances differ, the way forward - thoughtful design paired with real testing - remains useful far beyond this case.

References

- Aggarwal, C.C. (2018) *Machine Learning for Data Mining*. 2nd edn. Cham: Springer.
- Han, J., Kamber, M. and Pei, J. (2020) *Data Mining: Concepts and Techniques*. 4th edn. Cambridge, MA: Morgan Kaufmann.
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021) *An Introduction to Statistical Learning: With Applications in R*. 2nd edn. New York: Springer.
- Laudon, K.C. and Traver, C.G. (2021) *E-Commerce: Business, Technology, Society*. 16th edn. Harlow: Pearson Education.
- Russell, S. and Norvig, P. (2021) *Artificial Intelligence: A Modern Approach*. 4th edn. Harlow: Pearson Education.
- Sakar, C.O., Polat, S.O., Katircioglu, M. and Kastro, Y. (2019) 'Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks', *Neural Computing and Applications*, 31(10), pp. 6893–6908. <https://doi.org/10.1007/s00521-018-3523-0>
- UCI Machine Learning Repository (2019) *Online Shoppers Purchasing Intention Dataset*. Available at: <https://archive.ics.uci.edu/ml/datasets/Online+Shoppers+Purchasing+Intention> (Accessed: 27 January 2026).
- Bishop, C.M. (2006) *Pattern Recognition and Machine Learning*. New York: Springer.
- Mitchell, T.M. (1997) *Machine Learning*. New York: McGraw-Hill.
- Hastie, T., Tibshirani, R. and Friedman, J. (2017) *The Elements of Statistical Learning*. 2nd edn. New York: Springer.